

Local Credit, Global Settlement

Enhancing Trust-Based Credit Creation with On-Chain Reserve Contracts

Alexander Chepurnoy

Ergo

8th CCCS Conference — RAMICS

Plurality of Social Currencies: solidarity economy, municipal and community currencies for inclusive and sustainable futures

Rio de Janeiro, Brazil, June 8–12, 2026

The Need for Monetary Expansion

Blockchain Assets Are Inelastic

- Bitcoin, Ergo, and similar assets have **algorithmically predetermined emission schedules**
- Supply **cannot expand or contract** in response to real-world economic demand
- Entirely **disconnected** from productive activity, trade volumes, or credit needs

ChainCash: On-Chain Notes with Collective Backing

- Peer-to-peer money creation through trust and/or collateral
- Notes backed by **all previous spenders** in the chain
- Redemption against **any reserve** in spending history, with re-redemption later possible against an earlier one
- Fully on-chain — **transparency and censorship resistance**

Limitation

- Every transaction recorded on the blockchain
- Scalability and cost issues for frequent use
- Hard to apply in **community currency** setting with many small payments

Offchain Payments Today: No Monetary Expansion

Existing Bitcoin Layer-2 Systems

System	Mechanism	Collateral Requirement
Lightning Network	Bidirectional payment channels	100% locked on-chain
Cashu	Chaumian blind signatures (e-cash)	100% mint reserves
Fedimint	Federated threshold custody	100% federation backing

The Problem: No Credit Expansion

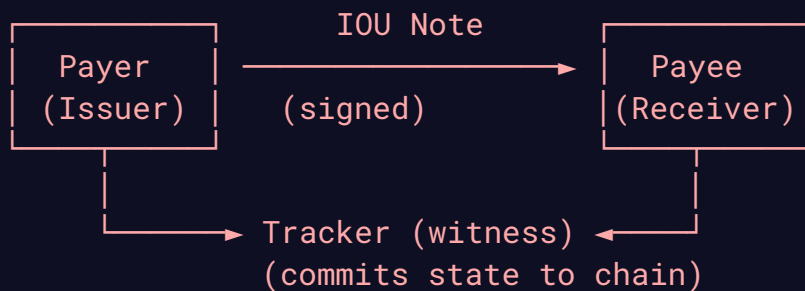
- Supply is capped by locked collateral
- Cannot respond to growing economic activity

Contrast with Community Currencies

- Community currencies (LETS, Time Banks, WIR) thrive on **trust-based credit**
- Existing digital systems **discard this principle** by enforcing full collateralization
- **Trust within communities becomes an underutilized resource**

Basis: A Protocol to unify Community Credit and Blockchain Assets

- **Local trust-based IOUs** circulate within communities
- **On-chain reserves** serve as optional backing
- Individual IOU note acceptance
- **Off-chain payments** — low fees, no blockchain bottleneck
- **Tracker service** coordinates debt state transparently



Note Creation: Issuing Community Credit

1. **Payer** (e.g., a community member) creates an IOU note:
 - Payer's **signature**
 - Payee's **public key**
 - **Debt amount**
 - **Timestamp**
 - Optional: **reserve contract** for extra backing
2. **Payee** verifies payer's signature and note contents
 - **Acceptance is always voluntary** — each participant decides whom to trust
3. **Tracker signature** is obtained later
 - Required only for **redemption** against a reserve contract
 - Certifies the debt was recorded in tracker's authenticated state

Note Redemption: From Trust to Settlement

Standard Redemption (Trust + Collateral)

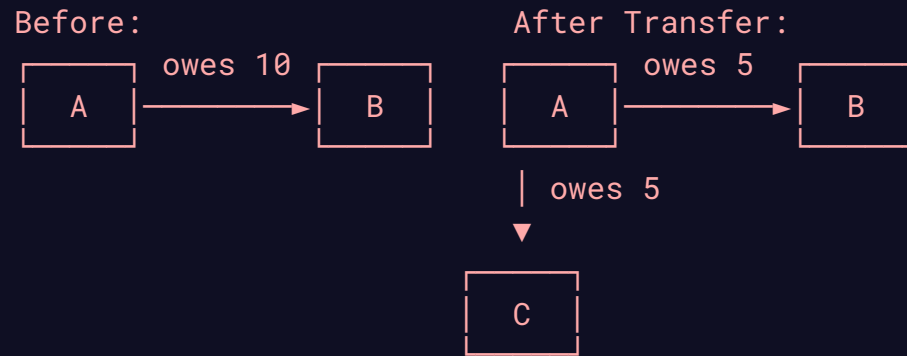
- Payee presents the **witnessed IOU** to the **reserve contract**
- Contract verifies payer's signature, tracker's signature, debt amount
- Redeemed amount is **released to payee** by the reserve contract

Emergency Exit (Tracker Unavailable)

- After **3 days (2,160 blocks)**, tracker signature becomes **optional**
- Redeem using payer's signature + last committed tracker state
- **Replay attacks prevented** by timestamp verification

Debt Transfer: Circulating Credit Within Communities

Triangular Trade (No Blockchain Needed)



- **B** requests **A** to co-sign:
 - Reduced debt $A \rightarrow B$
 - New debt $A \rightarrow C$
- Tracker witnesses both; old note replaced
- **A now owes C directly**

Tracker State: Transparent Debt Registry

AVL Tree of Debt Relationships

$\text{hash}(\text{payerKey} || \text{payeeKey}) \rightarrow (\text{totalDebt}, \text{timestamp})$

- Cryptographically strong hash function
- Efficient lookups and **proof generation**
- Root digest **committed to blockchain periodically**
- Enables **emergency redemption** and **public verification**

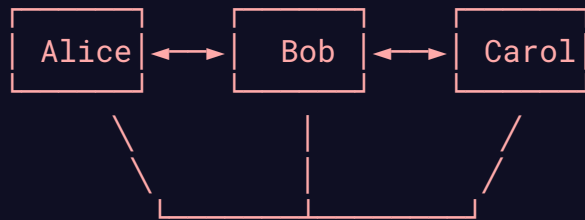
Governance Options

Design	Trust Model	Best For
Single Server	Trusted operator	Small communities
Federation (MuSig2)	Threshold of signers	Municipal currencies
Sidechain (PoW)	Permissionless consensus	Regional / multi-community

Scenario 1: Rural Community Trading

Disconnected Village with Occasional Connectivity

Village (no direct Internet)



Local Tracker
(syncs via satellite/cellular)

- Trade on **credit** within community over **Bluetooth / LoRa**
- Tracker commits to blockchain **when Internet available**
- Redemption possible via **slow links (SMS, email)**
- **No forced collateralization** — trust is enough locally

Scenario 2: Municipal Currency with Gradual Backing

A City-Scale Complementary Currency

1. **Phase 1 — Pure Trust:** Local businesses issue IOUs to each other
 - No collateral required
 - Tracker (city federation) witnesses transactions
2. **Phase 2 — Selective Backing:** As trade expands to outsiders
 - Businesses optionally create **on-chain reserves**
 - Reserves back IOUs for **inter-city / low-trust** transactions
3. **Phase 3 — Hybrid Economy:** Local trust dominates internally, reserves handle external settlement

Security & Trust Minimization

Tracker Cannot Steal Funds

- Tracker only **certifies inclusion** in state
- **Issuer's signature always required** for redemption
- Tracker cannot unilaterally create or redeem notes

Anti-Censorship

- Pseudonymity allows picking new keys
- Reserve contract has **refund option** (long timeout)
- Future: blind signatures + zero-knowledge proofs

Emergency Exit

- After timeout: tracker signature **optional**
- Same message format: `key || totalDebt || timestamp`
- Reserve owner's signature still required

Implementation

- Ergo blockchain as it has all the needed functionalities and superior for offchain protocols
- Supports ERG as well as custom tokens (stablecoins etc) for reserves
- Offchain debt tracker

Resources

- Paper & code: github.com/ChainCashLabs/chaincash
- Whitepaper: <https://github.com/BetterMoneyLabs/chaincash/blob/master/docs/basis/basis.pdf>
- Tracker: <https://github.com/BetterMoneyLabs/basis-tracker>
- Everything is open sourced and under public domain
- Contributions are much needed

Thank You

Alexander Chepurnoy

Ergo

kushti@protonmail.ch

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